

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Validation & Test: Hardware- and software-in-the-loop, regression suites, replay harnesses, parameter sweeps, pytest

Test Automation: Python test tooling, build and release pipelines, GitHub Actions, headless batch runs, Docker

Flight & Robotics: PX4 flight stack, flight testing, visual-inertial odometry, IMU sensor fusion, state estimation, telemetry logs

Debug: Diagnostics firmware, CAN/DBC, bus trace analysis, latency measurement, GDB, BusMaster, MoTeC i2 Pro

Languages: Python, C/C++, Java, SQL

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Leading a model-, software-, and hardware-in-the-loop effort building battery, motor, and inverter plant models so control code is validated against a simulated vehicle before it reaches hardware
- Set code standards and an integration plan for a modeling team of **7 senior engineers** so independently developed subsystem models compose into a system-level testbench
- Standing up a build and release pipeline for control-unit firmware, specifying the compute, storage, and CI infrastructure it needs

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed embedded flight-control changes in C and iterated gains through repeatable flight tests and log review
- Integrated and tuned visual-inertial odometry for onboard state estimation and evaluated performance against logged telemetry
- Tuned PX4 PID flight-control loops to reduce vertical and horizontal drift during autonomous position hold

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Built Python telemetry parsers and a software-in-the-loop replay harness to validate controller changes against recorded logs before hardware testing
- Planned and executed on-vehicle tests, measured response and latency, and produced plots for regression review
- Built diagnostics firmware exposing live power and torque frames, status bits, and error counters during test

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Extracted a desktop simulator and regression suite to reproduce field scenarios, run controller sweeps, and prevent regressions
- **Results:** 1st in endurance at MIS EV 2025

PROJECT EXPERIENCE

SR-Wireless-CAN — Telemetry Capture, Replay & Flashing Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, bus trace analysis, FastAPI, WebSockets, SQLite, Docker

- Added replay of recorded bus traces plus simulated and hardware capture modes, so test runs could be reproduced and re-examined off the vehicle
- Proposed and implemented wireless firmware flashing for the control unit, decoding an undocumented flashing sequence from bus trace captures and reimplementing it in Python

Vehicle Dynamics Simulation

Jul 2025

Spartan Racing, Formula SAE | San Jose, CA

Python, Pygame, Matplotlib, vehicle dynamics, PID control, CSV logging

- Built a physics-based simulator for torque, drag, acceleration, and steering to evaluate control changes before committing them to hardware
- Added headless batch runs, seeded scenarios, CSV summaries, and plots for repeatable offline comparison between builds